e-Government Procurement Implementation in India: Two Comparative Case Studies from the Field

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ABSTRACT

The turn of century saw ICT technologies making inroads into our lives. The governments world over are trying to use this medium for reaching out to their citizenry. This migration has been partly driven by transparency, efficiency and wider-access related benefits accrued by automation of government functions. E-Government Procurement (E-GP) is one such area which has drawn attention of politicians and researchers equally. However, studies bring out that E-GP project like any other e-Governance project has 70% chances of failure. Studies also underline that success of any E-GP system among other things, is affected by national culture, project environment (E-Readiness, IT literacy level and technological evolution) and regulatory framework which govern public procurement in a country. Indian government having realized the transparency and efficiency related benefits of E-GP system initiated an aggressive campaign to ensure its speedier implementation through National E-Governance Programme (NeGP) which was launched in 2006. However, recent review by Government of India (GoI) brought out that E-GP usage in the country has been less than satisfactory. In above perspective, this paper aims to undertake Template Analysis of stage-wise importance of 11 Critical Success Factors (CSF) (reported in literature) in E-GP project outcome in Indian context. For covering large landscape of E-GP implementation in India, two representative systems i.e. those implemented by Government of Andhra Pradesh and National Informatics Corporation were selected. The study concludes that out of 11 CSFs, five CSFs are not important at stage 2 of E-GP project evolution, while all 11 CSFs contribute to E-GP project success at Stage 3 and 4.

Categories and Subject Descriptors

J.1 **[Computer Applications]:** Administrative Data Processing: Business, Government; K.4.1 [**Public Policy Issues**]: Ethics, Regulation, Transborder data flow, Use/abuse of power; K.4.3 **[Computers and Society]:** Organizational Impacts: Automation, Computer-supported collaborative work

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General Terms

Economics, Human Factors, Legal Aspects, Management, Performance, Standardization, Theory

Keywords

Public Procurement; E-Procurement; Template Analysis; Case Study; Public Procurement Law

1. INTRODUCTION

The E-GP benefits related to transparency, efficiency and wider-access has also been brought out in other trans-national studies [14]. GoI having realized these benefits made E-GP a key result area for achieving efficiency in provisioning of goods / services which constitutes 30% of Indian Union Budget or 3.4% to 5.7% of Indian GDP [19, 10]. These efforts are further reinforced by the findings of E-GP case studies (conducted in India) which report savings of up to 25% over the procurement carried out through traditional manual mode [31, 22]. However, studies also underline that E-GP like any e-Governance Project has 70% chances of failure [18, 35]. In order to avert pitfalls which could lead to failure of E-GP in India, central government made it an Integrated Mission Mode Project (MMP) under the National e-Governance Plan (NeGP) launched in 2006. Designation of E-GP as an integrated MMP provides necessary organisational thrust to the project and institutional monitoring of the progress. The objective of E-GP MMP is "to create a national initiative to implement procurement reforms, through the use of electronic Government procurement, so as to make public procurement in all Sectors more transparent and efficient". However, the progress of E-GP implementation and its subsequent usage in the country has been low [10, 24].

Since each E-GP implementation has its own environment context, studies conducted in other countries are not very relevant to Indian context. Therefore, it was imperative to carry out India specific study to understand the factors which affect E-GP project outcome. While the empirical study has its own advantages, the questionnaire method is too simplistic to fully capture dynamics of e-government project especially E-GP. Therefore, case study method is popular for among the scholars with 48% studies conducted in E-Procurement area during 1997-2007 have either followed case study approach or employed content analysis of secondary data/ e-procurement websites etc [30].

2. LITERATURE REVIEW

The term E-Procurement has been defined by many authors from different perspectives [23]. While some scholars emphasize evolutionary nature of migration of procurement functions [24] to Internet-based system that offers electronic purchase, order processing and enhanced administrative functions to buyers, suppliers [21] and management [1]. Other scholars define e-procurement as ‘using Internet technology in the purchasing process’ [4, p. 26]. This divergence and use of plethora of terms defining e-procurement has been underlined by Schoenherr and Tummala [30] in their studies. They recommend that contextual definition for e-procurement should be developed and used for each study. This contextual definition provides the backdrop in which the e-procurement related study has been conducted and avoids confusion. Therefore, we have used the following custom definition for this study:

“Evolutionary migration of all manual procurement related functions (being undertaken by an organisation) to ICT platform(s) for documented objectives after undertaking procurement process re-engineering necessary for their e-adoption” [24, p. 5].

The importance of documenting e-procurement success factors in experience sharing has been brought out by Davila, Gupta, and Palmer [7]. However, it was Rockart [29] who first used critical success factors (CSF) in the context of information systems and project management. Further, credit of empirically linking CSFs to project success goes to Pinto and Slevin [26]. They defined a model of a project implementation success as S = f (X1, X2,......,Xn) where S is project success and Xi the critical success factor i. Needless to say that identification and communication of CSFs ensures that everyone in the project team is focused and thereby ensures project success. However, there has been divergence in scholarly views on measures related to project success. Despite this, majority of scholars tend to agree with the definition provided by Baker, Murphy, and Fisher [2] [25, 27]. Therefore, in this study the criterion propounded by Baker et al. [2] has been used. These measures of project success were considered while selecting candidate cases for the study as well while carrying out stakeholders’ interview. Firstly, by employing the degree of criticality test [37], 11 Critical Success Factors (CSF) of E-Procurement were identified during extensive literature review. The factors considered both necessary for and associated with project success have been selected as a CSF/Sub-Factor. These CSFs and their supporting studies are covered in Table 1 below.

Table 1: Critical Success Factors and their Sub-Factors

CSF/ Construct		Supporting Studies
1	**Top Management Support (TMS)**	
	Understanding of capabilities and limitations of IT	Finnegan and Golden (1996), Somers and Nelson (2001), Eid et al. (2004), Ngai et al. (2004), Ranganathan et al. (2004), Jeon et al. (2006)
	Approval from top management	Parida and Parida (2005), Dooley and Purchase (2006), Gunasekaran and Ngai (2008), Kaliannan et al. (2009), Panda and Sahu (2011)
	Identification of the project as top priority	Nah et al. (2001), Nah et al. (2003), Vakola and Wilson (2004), Loh and Koh (2004), Plant and Willcocks (2007)
	Alignment of system with business strategy	Ho and Pardo (2004), Dooley and Purchase (2006), Kampstra et al. (2006), Bricknall et al.(2007)
	Allocation of appropriate resources	Vakola and Wilson (2004), Croom and Brandon-Jones (2005), Gunasekaran and Ngai (2008), Kaliannan et al. (2009)
	Establishment of appropriate work culture	Kampstra et al. (2006), Kim (2006), Dooley and Purchase (2006), Lee et al. (2008), Engstrom et al. (2009), Kaliannan et al. (2009), Panda and Sahu (2011)
2	**E-Procurement Implementation Strategy (PIS)**	
	Sound procurement practices	Carpineti et al.(2006), Bof and Previtali (2007) and (2010), Coscia and Rubattino (2008), Hardy and Williams (2008)
	Opportunities for aggregation	Angeles and Nath (2005), Croom and Brandon (2005), Dooley and Purchase (2006), Batenburg (2007), Chen et al. (2009), Panda and Sahu (2011), Khanapuri et al. (2011)
	Relationships with industry and small businesses	Dooley and Purchase (2006), Al-Omoush (2008), Amelinckx et al. (2008), Bof and Previtali (2010), Chang and Wong (2010), Panda and Sahu (2011)
3	**Business Case and Project Management (BCPM)**	
	Identification of business drivers	Benslimane (2004), Lian and Laing (2004), Kaliannan et al. (2009), Ronchi et al. (2010), Sahu et al. (2010)
	Business process assessment	Santema and Reunis (2003), Vaidya et al. (2004a), Engström et al. (2009), Kaliannan et al. (2009), Jha et al. (2010)
	Return on investment (ROI)	Koorn et al. (2001), Capgemini (2004), Brandon-Jones (2009), Kelkar (2010), Khanapuri et al. (2011)
	Total cost of ownership (TCO)	CeG (2007), Al-Omoush (2008), Amelinckx et al. (2008), Martínez-Martínez (2008), Lou and Alshawi (2009)
	Risks identification and mgmt	Croom and Brandon-Jones (2005), Parida and Parida (2005), Schapper et al. (2006), Foerstl et al. (2010)
	Pilot projects	Ramsay and Croom (2008), Lee et al. (2008), Panda and Sahu (2011)
4	**Business Process Re-engineering (BPR)**	
	Diagnostic and analysis of processes	Angeles and Nath (2005), Bevilacqua et al. (2005), Trkman and Groznik (2006), Bof and Previtali (2007), Plant and Willcocks (2007)
	Redesign of business processes	Somers and Nelson (2001), Angeles and Nath (2005), Trkman and Groznik (2006), Bof and Previtali (2007), Lee et al. (2008), Khanapuri et al. (2011)
	Contributions of IS managers	King (2001), Galliers and Leidner (2003), Lai and Mahapatra (2004), Paper and Chang (2005), Bof and Previtali (2007)
	Integrated supply chain processes	Angeles and Nath (2005), Auramo et al. (2005), Defee and Stank (2005), Sanders and Premus (2005), Simatupang and Sridharan (2005)
	Collaborative process improvement	Skjoett et al. (2003), Angeles and Nath (2005), Lambert et al. (2005), Simatupang and Sridharan (2005), Tummala et al. (2006), Bof and Previtali (2007)
	Continuous process	Panayiotou et al. (2004), Beresford et al. (2005), Defee and Stank (2005),

CSF/ Construct		Supporting Studies
	improvement	Tummala et al. (2006), Trkman and Groznik (2006), Panda and Sahu (2011)
5	**Technology Standards (TS)**	
	Technical standards	Cimander et al. (2009), Engstrom et al. (2009), Bikshapathi et al. (2010), Panda and Sahu (2011)
	Content standards	ADB (2004), Döring et al. (2006), Engstrom et al. (2009)
	Process and procedural standards	Somasundaram and Damsgaard (2005), Gunasekaran and Ngai (2008), Kaliannan et al. (2009), Panda and Sahu (2011)
	Compliance with standard frameworks	Panda and Sahu (2011), Cimander et al. (2009), Engstrom et al. (2009), Gunasekaran and Ngai (2008), Kaliannan et al. (2009), Khanapuri et al. (2011)
	Interoperability	ADB(2004), Kaliannan et al.(2009), Panda and Sahu (2011)
6	**Security and Authentication (SA)**	
	Software, hardware availability and reliability	Ngai et al. (2004), Loh and Koh (2004), Bof and Previtali (2007), Moon (2005), Panda and Sahu (2011)
	Web-based systems' reliability	Soliman and Janz (2004), Gunasekaran and Ngai (2004a), Goo and Nam (2007), Wong et al. (2007)
	Web-based systems' scalability	Ball et al. (2002), Galliers and Leidner (2003), Soliman and Janz (2004), Wong et al. (2007)
	Security of Web-based systems	Moon (2005), Parida and Parida (2005), Gunasekaran and Ngai (2008), Cimander et al. (2009), Panda and Sahu (2011)
	IT previous experiences	Morgan-Thomas and Bridgewater (2004), Gunasekaran and Ngai (2004a), Zou and Seo (2006)
7	**System Integration (SI)**	
	Supportive cultural environment	McIvor et al. (2003), Presutti (2003), Vakola and Wilson (2004), Angeles and Nath (2005), Moon (2005), Parida and Parida (2005), Li et al. (2007)
	Information accuracy	Simatupang et al. (2004), Cooper and Tracey (2005), Blackhurst et al. (2006), Tsai et al. (2005)
	Real-time information sharing	Balasubramanian and Tewary (2005), Tsai et al. (2005), Blackhurst et al. (2006), Cimander et al. (2009)
	Reliability of information	Kim and Im (2002), Weippert et al. (2002), Simatupang et al. (2004), Auramo et al. (2005), Wei and Wang (2007)
	Sharing of information with stakeholders.	Sherer (2003), Soliman and Janz (2004), Angeles and Nath (2005), Bagchi and Skjoett (2005), Cooper and Tracey (2005), Dooley and Purchase (2006)
	Information accessibility	Angeles and Nath (2005), Boyer and Hult (2005), Defee and Stank (2005), Peansupap and Walker (2005)
8	**Change Management (CM)**	
	Identification and management of key stakeholders	Bof and Previtali (2007), Lee et al. (2008), Kaliannan et al. (2009), Klafft (2009), Bhatnagar and Singh (2010), Schooner et al. (2008), Khanapuri et al. (2011), Panda and Sahu (2011)

CSF/ Construct		Supporting Studies
	e-Procurement impact assessment	Al-Omoush (2008), Aslani et al. (2008), Brandon-Jones (2009), Kaliannan et al. (2009), Bof and Previtali (2010), Bhatnagar and Singh (2010)
	Identification and mitigation of potential barriers	Carayannis and Popescu (2005), Kshetri (2007), Bof and Previtali (2007), Gunasekaran and Ngai (2008), Van (2009), Panda and Sahu (2011)
	Effective management of organizational resistance	Parida and Parida (2005), Kamann (2007), Bof and Previtali (2007), Al-Omoush (2008), Lee et al. (2008), Kaliannan (2009), Panda and Sahu (2011), Khanpuri et al. (2011)
9	**Performance Measurement (PM)**	
	Sharing a clear understanding of objectives	Soliman et al. (2001), Kanji (2002), Umble et al. (2003), Simatupang and Sridharan (2005), Kaliannan et al. (2009)
	Measurement of performance	Loh and Koh (2004), Panayiotou et al. (2004), Croom and Brandon (2005), Tummala et al. (2006), Lee et al. (2008)
	Identification of measurable performance indicators	Soliman et al. (2001), Mason and Lefrere (2003), Loh and Koh (2004), Panayiotou et al. (2004), Frolick and Ariyachandra (2006), Tummala et al. (2006)
	Alignment of compensation with performance evaluation	Stank et al. (2001), Umble et al. (2003), Panayiotou et al. (2004), Defee and Stank (2005), Min and Galle (2003), Simatupang and Sridharan (2005a), Lee et al. (2008)
	Guiding the stakeholders to improve overall performance	Panayiotou et al.(2004), Loh and Koh (2004), Simatupang and Sridharan (2005a), Jonsson and Gunnarsson (2005), Angeles and Nath (2005), Kaliannan et al. (2009)
10	**Training and Education (TE)**	
	Training and learning how to operate new IT tools	Somers and Nelson (2001), Eid et al. (2004), Umble et al. (2003), Dowlatshahi (2005), Loh and Koh (2004), Engstrom et al. (2009), Gunasekaran and Ngai (2008)
	Understand how system will change business processes	Somers and Nelson (2001), Nah et al. (2001), Presutti (2003), Loh and Koh (2004), Yang et al. (2006)
	Investment in knowledge capital	Dowlatshahi (2005), Zou and Seo (2006), Angeles and Nath (2005), Engstrom et al. (2009), Gunasekaran and Ngai (2008), Khanapuri et al. (2011)
	Supportive environment	Peansupap and Walker (2005), Zou and Seo (2006),Gunasekaran and Ngai (2008), Lee et al. (2008), Engstrom et al. (2009), Panda and Sahu (2011)
	Developing own in-house training	Ramamurthy et al.(1999), Nix et al. (2005), Ngai et al.(2004), Zou and Seo (2006)
	Continuous learning and training	Somers and Nelson (2001), Eid et al. (2004), Panayiotou et al. (2004), Zou and Seo (2006), Gunasekaran and Ngai (2008), Lee et al. (2008)
11	**Adoption by Stakeholders (AS)**	
	Reliability of stakeholders	Sherer (2003), Hsieh and Hiang (2004), Lippert and Forman (2005), Ibrahim and

CSF/ Construct		Supporting Studies
		Ribbers (2006), Tummala et al. (2006)
	Competence and compatibility of partners	Dooley and Purchase (2006), Ibrahim and Ribbers (2006), Angeles and Nath (2007), Bof and Previtali (2007), Kaliannan et al. (2009), Chang and Wong (2010)
	Vulnerability to additional risks	Balasubramanian and Tewary (2005), Lin et al. (2005), Yeh (2005), O'reilly and Finnegan (2005)
	Continuous communication	Croom and Brandon (2005), Defee and Stank (2005), Yeh (2005), Chu and Fang (2006), Engstrom et al. (2009).
	Procedures to monitor participant performance	Angeles and Nath (2005), Balasubramanian and Tewary (2005), Lippert and Forman (2005), Skjoett et al. (2005), Yeh (2005)
	Perceived satisfaction	Parida and Parida (2005), Wong et al. (2005), Chu and Fang (2006), Tummala et al. (2006), Kaliannan et al. (2009)

Adapted from Panda and Sahu [23]

As brought out earlier, this study focuses on the evolutionary nature of the E-GP systems. However, during literature review, it emerged that the system evolution models vary from one another in terms of number of stages and domain related contexts. After comparing various evolution models, the model proposed by Layne and Lee [17] was found most suitable for our study due to its simplicity and relevance to E-GP domain. The abridged version of this model has been previously used for similar study [28]. More so, the model has been recommended in India context [13]. The model [17] suggests that any E-Government application like E-GP evolves through four stages namely:

(a) Stage 1 (Catalogue): Government entity has a web presence for dissemination of information;

(b) Stage 2 (Transaction): Entails a working database which supports transactions.

(c) Stage 3 (Vertical Integration): Upstream and downstream integration of procurement functions within an E-GP system and across other E-GP systems takes place.

(d) Stage 4 (Horizontal Integration): E-GP system gets integrated with other associated systems like financial applications etc.

3. RESEARCH OBJECTIVES

Importance of national culture of a country [3, 33], its public procurement policies as well as project environment [34, 15] has been underlined by many researchers. Further, it has been stated that due to differences in enabling conditions like IT infrastructure, penetration of Internet and IT literacy [12, 32], existing E-GP studies have limited applicability to India [22]. Therefore, objectives for this study were to ascertain stage-wise importance of CSFs (reported in literature) in E-GP project outcome in Indian context. Further, the study aimed to provide few recommendations based on analysis of the project management experience in the candidate E-GP systems.

4. RESEARCH DESIGN

The research hypothesis for the study was adapted from Pinto and Slevin [26]. Accordingly, it was hypothesized that E-GP 'Project Success' (PS) is a function of 11 CSFs identified during literature review i.e. PS = f (TMS, PIS, BCPM, BPR, TS, SA, SI, CM, PM, TE, AS) where PS is project success for each Stage of E-GP evolution. For fine-tuning our research model, we used the findings of exploratory study carried out by Panda and Sahu [24] in Indian context.

4.1 Status of E-GP Implementation in India

All states and Union Territories in India except Andaman & Nicobar have migrated to E-GP post issuance of Government of India directive in 2010. Even Andaman & Nicobar is in the process of implementing E-GP system. However, usage of E-GP system varies widely from one implementation to another. Moreover, E-GP implementations in India are at Stage 2 of project evolution [24].

One of the interesting finding of the study by Panda and Sahu [24] is that the e-procurement implementation in India has been either through Public Private Partnership (PPP) or the National Informatics Cente (NIC) route. NIC is a government of India undertaking which provides IT services to various government departments. The NIC has been entrusted by GoI with the responsibility of providing many key services which would ultimately form the backbone of NeGP. While early E-GP implementers like states of Andhra Pradesh, Chhattisgarh, Gujrat, Karnataka etc., have followed PPP route, the remaining state governments have implemented NIC e-Procurement software solution (GePNIC). In addition, many state governments like Assam, Punjab etc., who were earlier using PPP solutions have migrated to GePNIC in last couple of years. This general preference for GePNIC could be attributed to following benefits (perceived by state governments) of adopting GePNIC:

(a) The solution is audited and certified by STQC (a GoI agency dealing with testing and quality certification of E-Gov systems). Therefore, use of GePNIC would save them the hassle of getting the system audited.

(b) The NIC also undertakes the role of Application Service Provider, hence the burden of hosting and managing the security of the online portal lies with NIC. This could be one of the reasons as to why most states who were late in E-GP implementation or low on e-Readiness Index preferred GePNIC over PPP/ in-house development. Moreover, implementing GePNIC did not entail tendering or cost since NIC is funded by GoI.

(c) The interfacing of the GePNIC solution with other higher level systems such as banking gateways is presumed to be easier in comparison to other routes. This is because NIC is a GoI undertaking with requisite inroads into bureaucratic red-tapism.

(d) The upstream linking of the portal with National Service Delivery Gateway (NSDG) would be seamless as NSDG is operated by NIC.

4.2 Research Hypothesis

Panda and Sahu [24] brought out that E-GP systems in India are at Stage 2 of evolution with activities related to Stage 3 of project evolution being attempted. Further, since Stage 1 of E-GP evolution is too trivial, it was not considered for the study. In view of the foregoing, it was felt necessary to have two sets of hypotheses, H1 to capture the actual experience of respondents for the Stage 2 of E-GP project evolution and H2 for capturing opinion of the respondents about the importance of CSF in Stage 3 & 4 of project evolution. Accordingly, two sets of hypotheses sets i.e. H1 (for Stage 2) and H2 (for Stage 3&4) were formulated for stage-wise testing of project success based on the Pinto and Slevin [26] equation of project success. It was proposed to validate the equation indirectly through 11 sub-hypotheses with each sub-hypothesis representing one CSF. These hypotheses sets are covered in Table 2.

Table 2: Hypotheses for the Study

Hypothesis	Statement
For Stage 2 of E-GP Evolution: PS = f (TMS, PIS, BCPM, BPR, TS, SA, SI, CM, PM, TE, AS) i.e.	
Hypothesis Set 1 (H1)	**H1.1**: TMS is a predictor of E-GP Project Success
	H1.2: PIS is a predictor of E-GP Project Success
	H1.3:BCPM is a predictor of E-GP Project Success
	H1.4: BPR is a predictor of E-GP Project Success
	H1.5: TS is a predictor of E-GP Project Success
	H1.6: SA is a predictor of E-GP Project Success
	H1.7: SI is a predictor of E-GP Project Success
	H1.8: CM is a predictor of E-GP Project Success
	H1.9: PM is a predictor of E-GP Project Success
	H1.10: TE is a predictor of E-GP Project Success
	H1.11: AS is a predictor of E-GP Project Success
For Stage 3 & 4 of E-GP Evolution: PS= f (TMS, PIS, BCPM, BPR, TS, SA, SI, CM, PM, TE, AS) i.e.	
Hypothesis Set 2 (H2)	**H2.1**: TMS is a predictor of E-GP Project Success
	H2.2: PIS is a predictor of E-GP Project Success
	H2.3: BCPM is a predictor of E-GP Project Success
	H2.4: BPR is a predictor of E-GP Project Success
	H2.5: TS is a predictor of E-GP Project Success
	H2.6: SA is a predictor of E-GP Project Success
	H2.7: SI is a predictor of E-GP Project Success
	H2.8: CM is a predictor of E-GP Project Success
	H2.9: PM is a predictor of E-GP Project Success
	H2.10: TE is a predictor of E-GP Project Success
	H2.11: AS is a predictor of E-GP Project Success

4.3 Selection of the Cases for the Study

For selecting E-GP systems for the qualitative case study, the 'India: e-Readiness Assessment Report 2008' [9] was used. From the long list of states and E-GP solutions, E-GP implementation by Govt of Andhra Pradesh and GePNIC was selected. The rationale for selection of the cases is enumerated below:

(a) Andhra Pradesh figures in the leader position in e-Readiness rankings and favorable environment exists which would facilitate implementation of any e-procurement system. Moreover, the state was the first government entity to implement E-GP solution in the country in 2003. It faced many problems during its arduous journey entailing migration of manual procurement to E-GP system. Usage of the system (ascertained during exploratory survey) was another criterion which influenced the selection of Andhra Pradesh for case study. The lessons learnt by the state and the measures adopted by the state overcome them is very important from this research point of view.

(b) Exploratory study brought out that majority of E-GP implementations in the country (especially post 2011) have been through GePNIC route. The study also brought out that the utilization of the system varies widely from one implementation to another. The system (though cannot be deemed very successful), was included for the case study as a reference system to compare and contrast it with the successful PPP mode E-GP implementation by GoAP. It was expected that understanding developed by inclusion of the GePNIC in the case study would also help in evolution of pertinent recommendations for improvement of GePNIC implementations.

4.4 Methodology Adopted for Analysis

For undertaking qualitative analysis of the selected case studies and describing them in an unambiguous, logical and consistent way, Template analysis of 11 CSFs was carried out. Template analysis is the process of organizing and analyzing textual data according to themes. This can be text produced or used in the context of the evaluation irrespective of the evaluation activity. Sections of the text are described and organized according to themes considered to be important to the evaluation. This involves indexing sections of the text to be relevant to a particular theme. Unlike the quantitative approach of content analysis, developing the codes is part of the analysis process. These codes are known as 'template'. The coding structure tends to be hierarchical with sub-themes emerging within themes [6, 11, 16, 36]. The analytical framework used for our case study is shown in Table 3. The template provided a guide for conducting semi-structured interviews and screening of documents (e.g. documents, usage statistics, E-GP portal etc).

Table 3: Template used for case study

1.1	Motivation for migration to e-procurement
E-Procurement Implementation	
1.2	Top Management Support
1.3	E-Procurement Implementation Strategy
1.4	Business Case and Project Management
1.5	Business Process Re-engineering
Project Evolution	
1.6	Change Management
1.7	Training and Education
1.8	Adoption by Stakeholders
E-Procurement System Details	
1.9	Technology Standards
1.10	System Integration
1.11	Security and Authentication
1.12	Performance Measurement

4.5 Research Tool and Techniques

An in-depth analysis of the available literature and procurement policies of the state/ government organization concerned was carried out. The case study entailed semi-structured interview of stakeholders (E-GP System Users, E-GP System Implementing Agency, PPP partners and Suppliers) of E-GP system in selected State Govt/ NIC. In addition, in appropriate cases snow balling approach [8] and referral by the domain experts [38] were also employed to develop an adequate interview sample.

The recording of the interviews and associated documents were processed through NVivo 8. NVivo is potent software which enables systematic analysis of qualitative data using conceptual approach. The conceptual content analysis facility of NVivo enables visualization of occurrence of selected terms within a text or texts [5, 20].The recurrence of the key terms of the template by the interviewee indicates their strength or weakness. The statistical data/ documents acquired during the interviews were used for validating the averments made during the interview.

5. CASE 1: E-GP SYSTEM OF GOVT OF ANDHRA PRADESH

The Government of Andhra Pradesh (GoAP) took many e-Government initiatives under the leadership of the Chief Minister Sh N Chandrababu Naidu. One of such initiative was constitution of a Cabinet Sub-Committee in 2000. The Cabinet Sub-Committee examined various issues relating to revision and streamlining of tender procedures. The Cabinet Sub –Committee analyzed the deficiencies in the procedures (in vogue during that time) in respect of registration of contractors, qualification criteria, verification of certificates, standard bidding documents, tender premium, purchase of tender documents by non-serious bidders, prevention of cartel formation, deduction of taxes at source, maintenance of assets, grading of contractors and engineers, quality control measures etc., and examined different alternatives for achieving better results and made recommendations to the Government for consideration. One of the major recommendations of the committee was introduction of e-procurement system in the state.

The initiatives of Sh N Chandrababu Naidu were followed through by next Chief Minister. The system was implemented through PPP route with M/s C1 India as private partner. The cost of implementation to GoAP was nil as PPP partner is compensated by participating bidders by paying a stipulated percentage of the Estimated Contract Value (ECV) while uploading tender. In the ensuing years, this humble beginning by the state (being the first implementation of E-GP in India) saw many ups and downs. The implementation has received many accolades like UN Public Service Award in 2007 and Top 20 Best Innovations in the World conferred by IBM-Ash Institute, Harvard University in 2007. After expiry of initial terms of PPP contract, the solution was taken over by GoAP. M/s Vayam Technology Ltd has been awarded O&M contract for the system wef 2010. Digital signatures for use by stakeholders of the system are issued by APTS (a Govt of Andhra Pradesh Undertaking) who is sub Certifying Authority (CA) of M/s Tata Consulting Services.

6. CASE 2: GePNIC

National Informatics Centre (NIC) was established in 1977. Satellite based computer network NICNET was launched in 1987 which was later extended to state capitals. NIC started working on GePNIC after facing problems in using TMS solution provided by M/s C1 India who emerged as L1 in the competitive bidding undertaken by NIC for procuring E-GP services/ platform. Unsatisfied with the product, NIC launched a pilot project for its own solution in 2007-08 in Tamil Nadu. Subsequently, Central Public Procurement Portal was setup by NIC on directions of Apex Committee headed by Cabinet Secretary in 2010. The GePNIC solution has received many awards like E-India e-Gov Award in 2007 and Skotch Award in 2010. NIC extends GePNIC solution to willing states/UTs. No charges payable either by the Govt or the bidder to NIC. The operating cost of NIC is borne by Dept of Commerce, Ministry of Commerce & Industry, GoI. Tender charges / EMD related issues are dealt by the respective states/ UTs. IPR of the solution is with NIC. NIC is a CA and issues DS to all stakeholders.

7. TEMPLATE ANALYSIS OF THE CASES

The template analysis of the cases viz-a-viz 11 CSFs identified through extensive literature review is covered in table 4.

Table 4: Template analysis of cases

GoAP	GePNIC
Top Management Support	
The top leadership recognized potential of Information and Communication Technologies (ICT) in reducing corruption and improving efficiency.	GoI recognized potential of ICT technologies in reducing corruption and improving efficiency. E-GP is an integrated MMP under NeGP.
Patron- Chief Minister	Apex Committee- Cabinet Secretary, Govt of India
The project was accorded priority and required resources were allotted.	Project has received importance since 2010, but allocation of resources at state level lacking.
Steering Committee- Chaired by Chief Secy. Members include concerned Principal Secretary/ Secretaries, Heads of all participating Depts, rep of PPP Partner.	Empowered Committee- Chaired by Secretary (Commerce).Members include Secretary IT, Director General of Supplies& Disposal and NIC & other secretary
Project Implementation Committee was constituted	Core Group- Chaired by Joint Secretary (Commerce)
Nodal Officer: Secretary IT& Comm	Nodal Officer at state level: Designated by each state/ UT with state NIC officer as member.
GoAP	**GePNIC**
E-Procurement Implementation Strategy	
Nodal Agency: IT&C Department	Nodal Agency: Dept. of Commerce, GoI
Consultant: PricewaterCoopers	Consultant: National Institute of Smart Governance
Model: PPP (M/s C1 India). IT&C took over system in 2010 and handed over to next ASP operator M/s Vyam Tech	Model: Internal development of source code by NIC through hiring of programmers.
Demand aggregation implemented for the financial powers devolved to lower functionaries districts/ departments etc.	Demand aggregation not implemented for the financial powers devolved to lower functionaries districts/ departments etc.
General Financial Rules 2005 (GFR) provisions followed.	GFR provisions followed in the E-GP system.
Compensation Model: 0.03% of Estimated Contract Value (ECV) of tenders paid electronically by bidders at the time of bid submission. Cap of Rs.10,000/- for all works with ECV up to Rs.50 Cr, and Rs.25,000/- for works with ECV above Rs. 50 Crore.	Compensation Model: No cost to Government department opting for E-GP solution or supplier. Operating costs are borne by Ministry of Commerce & Industry.
GoAP	**GePNIC**
Business Case and Project Management	
In 2000, Cabinet sub-committee recommended E-GP adoption.	Apex Committee Mandate.
Pilot: During Jan 2003 in five departments	Pilot project during 2007-08 in state of Tamil Nadu.
Total cost of ownership concept not applicable due PPP mode of implementation.	Total cost of ownership concept not applicable as NIC is funded by Govt of India.
Transaction fee model. No cost	No cost to state/ UT.

GoAP	GePNIC
to Govt.	
Mandatory procurement above Rs 1.0 Cr through e-route during pilot phase. Roll out: July 2004.	For Central Public Procurement Portal (CPPP), Rs. 10Lakh ECV, e-publish mandatory. For state/ UT portals, as decided by them.
GoAP	**GePNIC**
Business Process Re-engineering	
GFR processes implemented in E-GP system. Administrative changes related to tendering carried out. IT managers were actively involved in system implementation.	GFR processes implemented in E-GP system. IT managers were actively involved in system implementation.
Centralized registration of suppliers	Centralized registration of vendor for CPPP site. In addition, each state site has its own registration of vendors.
Tender document fees to be paid online only at the time of bid submission. Viewing the tender online is free	Tender document is free for viewing and downloading.
Manual signature replaced by digital signature.	Manual signature replaced by digital signature.
Scanned copy of Earnest Money Deposit (EMD) document made admissible.	Scanned copy of EMD document made admissible.
Compulsory use of E-GP for ECV above Rs 1 Lakh for works, goods & services.	Compulsory hosting of tenders on site with ECV beyond a threshold value which varies from state to state.
Misc: Redesign bid forms and auto bid evaluation. Universal item codes	Misc: Hosting of tenders on CPPP site is mandatory for ECV above Rs. 10 Lakh.
GoAP	**GePNIC**
Technology Standards, Security & Authentication, System Integration	
Certified by STQC and third party	Certified by STQC and third party
E-mail and SMS gateway to share real time tender information/ progress with stakeholders.	E-mail and SMS gateway to share real time tender information/ progress with stakeholders.
System integration was facilitated through executive orders.	System integration was facilitated through executive orders.
Change Management	
E-Procurement made mandatory through executive orders.	E-Procurement made mandatory through executive orders
The role of top management in ensuring compliance. Normally looked after by a Project Implementation Committee.	Top driven by Department of Commerce, Govt of India.
All procurements beyond a threshold value were to be procured through e-mode after a stipulated date.	All procurements beyond a threshold value are to be published on CPPP portal after a stipulated date.
Impact assessment of E-GP system was carried out during initial stages. Not followed through.	No such assessment carried out nor planned.
GoAP	**GePNIC**
Performance Measurement	
Clear timelines and performance measurement outcomes were stipulated. Targets were assigned and monitored.	The timelines were specified for states to adopt E-GP system. The adoption status is being monitored by Dept of Commerce, GoI.
IT&C Department was designated to assist users in optimal utilization of the system.	NIC was designated to assist users in optimal utilization of GePNIC system.
Tender cycle time reduced from average 90-135 days to 32 days.	No formal study on these aspects has been carried out. However, monthly review of the MMP is carried out by Joint Secretary (Commerce) based on project management information system reports generated through the solution developed by NIC. Accordingly, official communication to concerned states/ UTs for improvement.
Cost savings to a tune of Rs 22000 Crore.	
Supplier participation increased from an average of 3 per tender in conventional mode to 4.5 in e-Procurement mode.	
Transaction fee payable by a bidder is lesser than tender fee charged in the manual tender system.	
GoAP	**GePNIC**
Training and Education	
Hands-on Training / Trainers programme for User Organisations.	Training on two modules: e-Publish and e-Procure.
Hands-on Training / Trainers programme for User Organisations.	Hands-on Training / Trainers programme for User Organisations.
Training manuals prepared and uploaded on website.	Training manuals prepared and uploaded on website.
Miscellaneous: Supplier registrations demo entailing Mock bids. Initially, 9500 + officers and 20,000 + Contractors/ Suppliers trained.	Miscellaneous: Training through webinar by sending an email request to cppp-nic@nic.in with the preferred slot for training.
GoAP	**GePNIC**
Adoption by Stakeholders	
Awareness sessions.	Awareness sessions.
No of suppliers registered on E-GP portal: 60000	Data not available. Registration not mandatory on CPPP site.
No of dept using: 27 Dept, 43 PSU & Corporations, 20 Universities and 135 Urban Local Bodies Total tenders:308253 Nos Total value: Rs. 351529 Crore Data up to 31st March 2014.	CPPP Portal: Total tenders: 1327 amounting Rs. 2128.19 Crore. States using GePNIC as MMP: Total tenders: 5727 Amounting Rs.11902Crore States using GePNIC as non-MMP: Total tenders: 7763 amounting Rs. 4410.80 Crore Data up to July 2013.
Helpdesk manned by PPP Partner. Log of all Help Desk calls maintained. Call disposal tracked.	Helpdesk created.
Use of traditional formats and terminologies. Stakeholders consulted on changes.	Use of traditional formats and terminologies.

8. FINDINGS

As brought out earlier, both E-GP systems are at Stage 2 of project evolution. However, efforts are on for upstream and downstream integration of other procurement functions such as requisition management etc in E-GP system. During discussions, in both cases, views of the stakeholders on importance of 11 CSFs were obtained. However, in the absence of management experience pertaining to Stage 3&4 of E-GP evolution, the findings of this study for these stages would need to be weighed accordingly. The support for the research hypotheses in the case study is summarized in Table 5. For Stage 2, hypotheses H1.1, H1.2, H1.4, H1.9, H1.10, H1.11 were supported during case study. However, during discussions in the case states, it was brought out that the BPR need to be paid more attention in future for simplification of procedure for e-adoption. According to the interviewees, this would entail two pronged approach: amendment to procurement provisions stipulated by Ministry of Finance, GoI, at the same time state governments would need to optimize procurement process flows.

The case study 1 [GoAP] made out a business case for migrating to E-GP. The state further set up project management committee to undertake pilot project, change management etc. However, in Case 2, the changeover to E-GP was mandated through an executive order. Therefore, H1.3 received partial support in case 1; however, the same was not supported by case study 2. Initially, the CSFs represented by H1.5, H1.6 and H1.7 were catered by GoAP in their implementation in 2003. However, post issuance of GoI standards, the states asked the PPP partners to get their solution audited by STQC. PPP partners have been directed to comply with the GoI stipulations in this regard. Therefore, despite of initial importance given to these predictors, the focus has now shifted from management to compliance. Hence, these hypotheses were not supported at Stage 2 of evolution. As regards, change management (H1.8), the case states and NIC has focused more on capacity building. All changeovers in GePNIC system were affected through executive orders and monitoring by empowered committees with the exception of Andhra Pradesh where change champions were selected in various departments. Therefore, the hypothesis was not supported at Stage 2. As regards importance of the CSFs in Stage 3&4 (representing hypotheses H2.1 to H2.11) are concerned, the interviewees opined that these aspects would certainly become important as system grows in both functionality and coverage. The cases are quite different in terms of project ownership and management, yet the cases underline that political will, structured methodology and effective management of E-GP project pays off. Comparison of these implementations helped in understanding the aspects in which these states managed the e-procurement CSFs which differentiated them (successful) from GePNIC (not very successful). The specific findings of case study viz-a-viz research hypotheses are covered in Table 5.

Table 5: Hypothesis Support in Case Study Findings.

Predictor Variable	**Stage 2 of E-GP Evolution**		**Stage 3&4 of E-GP Evolution**	
	Hypo-thesis	**Status**	**Hypo-thesis**	**Status**
Top Management Support	H1.1	Supported	H2.1	Supported
E-Procurement Implementation Strategy	H1.2	Supported	H2.2	Supported
Business Case and Project Management	H1.3	NS	H2.3	Supported
Business Process Re-engineering	H1.4	Supported	H2.4	Supported
Technology Standards	H1.5	NS	H2.5	Supported
Security and Authentication	H1.6	NS	H2.6	Supported
System Integration	H1.7	NS	H2.7	Supported
Change Management	H1.8	NS	H2.8	Supported
Performance Measurement	H1.9	Supported	H2.9	Supported
Training and Education	H1.10	Supported	H2.10	Supported
Adoption by Stakeholders	H1.11	Supported	H2.11	Supported

Therefore, for Stage 2 of E-GP project evolution; Project Success (PS) is a function (f) of ‘Top Management Support’ (TMS), ‘Adoption by Stakeholders’ (AS), ‘Performance Measurement’ (PM), ‘Training & Education’ (TE), ‘E-Procurement Implementation Strategy’ (PIS) and ‘Business Process Re-engineering’ (BPR) i.e.

PS= f (TMS, AS, PM, TE, PIS, BPR).

While for Stage 3&4; Project Success (PS) is a function (f) of ‘Top Management Support’ (TMS), ‘Business Process Reengineering’ (BPR), ‘E-Procurement Implementation Strategy’ (PIS), ‘Change Management’ (CM), ‘Adoption by Stakeholders’ (AS), ‘Business Case & Project Management’ (BCPM), ‘System Integration’ (SI), ‘Training & Education’ (TE),‘Technology Standards’ (TS), ‘Performance Measurement’ (PM), ‘Security & Authentication’ (SA) (as each CSF received Mean Score above 1) i.e.

PS=f (TMS, BPR, PIS, CM, AS, BCPM, SI, TE, TS, PM, SA).

During the study, following points were emphasized by the interviewees:

(a) Top Political leadership support is the key for E-GP system implementation and its subsequent usage.

(b) High level Project Implementation Committees ensure that problems encountered during the project implementation are resolved without any wastage of time.

(c) Business Process Re-engineering is necessary for E-GP system success. Non value adding processes must be purged.

(d) It is important to involve Stakeholders in system planning and design.

(e) Capacity building of stakeholders through workshops/ training camps is necessary to facilitate usage of E-GP system. Active Helpdesk must be established for hand holding of stakeholders.

(f) Public Private Partnership Model is preferable mode for fast and cost effective scaling up e-procurement.

9. RECOMMENDATIONS

Following recommendations are based on the views expressed by various stakeholders of E-GP system during the case studies:

Top Management Support. Due impetus to the project is necessary. For this following three core committees should be formed for overseeing project execution:

(a) Empowered Committee: to address procedural/ legal impediments and take important project decision),

(b) Interdepartmental Committee: to form common set of business logic to be implemented in the e-procurement system and achieve inter-departmental coordination,

(c) Project Committee: to liaise with the PPP Partner/ NIC.

E-Procurement Implementation Strategy. Reporting facilities of the E-GP systems should utilise consumption data to provide valuable inputs for pricing and budgeting of public purchases. These MIS reports should be used by rate contract finalizing agencies for concluding contracts for government procurement at most economical rates.

Business Case and Project Management. The study brought out that the CSF would be important for Stage 3 & 4 of E-GP evolution. In these stages, as the E-GP system gets increasingly integrated with other systems, pilot projects may be undertaken for identification of risks and their mitigation before system-wide scaling up of evolutionary changes.

Business Process Re-engineering. The BPR process should aim to simplify archaic public procurement procedures and reduce entry barriers for the potential suppliers. Towards this, centralized registration of vendors at national level could be considered. Each vendor may be provided with unique Vendor ID and Digital Signature after due verification of their credentials. This would not only preclude entry barriers but also result in procedural efficiency as vendors would not be required to re-register with each procuring agency separately.

Technology Standards and Security & Authentication. Presently, these aspects are ensured through regulatory compliances. However in stage 3 & 4 as the E-GP system would get integrated with non e-gov system like banking etc., it would require availability of secure interfaces for data interchange. Therefore, clearly laid down standards would ensure interoperability and compatibility of diverse systems.

System Integration. There is a need to implement central repository for E-GP data at national level to derive MIS reports for making better procurement decisions. The data spread over the country may be migrated to a National Data Centre. This would streamline data management while aiding data mining.

Change Management. Central aid to states/ UTs for various schemes like Right to Education, Health Mission, Employment Guarantee etc., may be made contingent upon spending it through e-mode. Tax collected through e-payment mode may be shared with respective states in transparent manner. This would encourage states to migrate to e-payment. Effective monitoring of the system usage is a must.

Performance Measurement. Institution of award for outstanding achievement in E-GP arena may be implemented. This would encourage govt employees and department to use E-GP systems. Task of performance measurement including performance audit of E-GP system and its usage may be assigned to a third party.

Adoption by Stakeholders and Training & Education. Capacity building of stakeholders may be augmented by creating a national institute dedicated to public procurement. The research institute may recommend measures to Government regarding optimization of public procurement and E-Procurement system interfaces. The institute may also act as interface between industry and the Govt.

Legislative Intervention in the form of National E-Procurement Act. Since the subject ‘Procurement’ is not listed under the schedule wherein state governments have exclusive rights. Therefore, Indian parliament has jurisdiction to make law on the subject for entire country. Government of India may consider enacting a National e-Procurement Act which would help in addressing all E-GP CSFs.

10. CONCLUDING REMARKS

Migration of public procurement to Internet has many benefits for a developing country like India. The country could ill afford not to migrate to E-GP system which could provide savings upto 25% in public purchases. However, like other E-Gov projects of NeGP, E-GP has also been making very slow progress in the country. In this context, the key contribution of this study has been in outlining role of Critical Success Factors in E-GP project outcome. The study highlights and compares two pertinent cases i.e. successful E-GP implementation by states of Andhra Pradesh and GePNIC. The recommendations of the study could help political leadership as well as E-GP project managers in avoiding the pitfalls by paying adequate attention to the key result areas of the project.

REFERENCES

[1] Atkinson, B. 2007. How to Get the Most Value from E-Procurement Tool. Retrieved October 20, 2011, from http://www.mbtmag.com/article/CA6498473.html.

[2] Baker, B. N., Murphy, D. C., and Fisher, D. 1988. Factors affecting project success. In D. I. Cleland and W. R. King (Eds.), Project Management Handbook 2nd Edition. New York: Van Nostrand Reinhold, pp. 902 – 909.

[3] Batenburg, R. 2007. E-procurement adoption by European firms: A quantitative analysis. Journal of Purchasing and Supply Management. 13, 3, 182-192.

[4] Boer, L.D., Harink, J. and Heijboer, G. 2002. A conceptual model for assessing the impact of electronic procurement. European Journal of Purchasing and Supply Management. 8, 1, 25-33.

[5] Buchanan, J. and Jones, M. L. 2010. The efficacy of utilising Nvivo for interview data from the electronic gaming industry in two jurisdictions. Review of Management Innovation & Creativity. 3, 5, 1-15.

[6] Crabtree, B.F. and Miller, W.L. (1999). Using codes and code manuals: a template organizing style of interpretation. In B.F. Crabtree and W.L. Miller (Eds.), Doing Qualitative Research, 2nd Edition. Thousand Oaks, California: Sage Publications.

[7] Davila, A., Gupta, M., and Palmer, R.J. 2003. Moving Procurement Systems to the Internet: The Adoption and Use of E-Procurement Technology Models. Research Paper Series, Stanford. Stanford University: Graduate School of Business.

[8] Denscombe, M. 2007. The Good Research Guide: For small-scale Social Research Projects 3rd Edition. Berkshire, England: Open University Press, McGraw-Hill House.

[9] DIT and NCAER. 2010. India: e-Readiness Assessment Report 2008. http://www.ncaer.org/downloads/annual reports/Annual Report_2008.pdf. Retrieved June 20, 2012.

[10] DIT. 2011. Saraansh: A compendium of Mission Mode Projects under NeGP. http://www.mit.gov.in. Retrieved Dec 10, 2012.

[11] Gibbs, G. (2007). Analyzing Qualitative Data. London: Sage Publications

[12] Gupta, M. and Narain, R. 2012. A study on usage of IT and its implications on e-procurement in Indian organisations. International Journal of Business Information Systems. 10, 2, 222-244.

[13] Gupta, M. P., and Jana, D. 2003. E-government evaluation: A framework and case study. Government Information Quarterly. 20, 2003, 365–387.

[14] Kaliannan, M., Awang, H., and Raman, M. 2009. Electronic Procurement: A Case Study of Malaysia's e-Perolehan Initiative. International Journal of Electronic Governance. 2, 2-3, 103 – 117.

[15] Kim, J. 2006. Determinants of B2B Adoption in South Korean Fishery Wholesale Firms: An Exploratory Study. Reviews in Fisheries Science. 14.

[16] King, N. (2012). Template Analysis. In G. Symon and C. Cassell (Eds.), Qualitative Organizational Research: Core Methods and Current Challenges. Sage Publications, London.

[17] Layne, K., and Lee, J. 2001. Developing fully functional e-government: A four stage model. Government Information Quarterly. 181, 122-136.

[18] Maio A.D., Caldwell, F., and Kao, L. 2000. E-Government Strategy: Cubing the Circle. Gartner Research Note.

[19] OECD. 2010. Report on Collusion and Corruption in Public Procurement in India. http://www.oecd.org/dataoecd/19/20 /44530808.pdf. Retrieved Jun 23, 2011.

[20] Palmquist, R. A. 2001. An overview of usability for the study of users' Web-based information retrieval behavior. Journal of Education for Library and Information Science. 42, 2, 123-136.

[21] Panayiotou N., Tatsiopoulos I., and Gayialis S. 2004. An E-Procurement System for Governmental Purchasing. International Journal of Production Economics. 90, 1, 79-102.

[22] Panda, P. and Sahu, G.P. 2011. E-Procurement Implementation: Comparative Study of Governments off Andhra Pradesh and Chhattisgarh. IUP Journal of Supply Chain Management. 8, 2, 34-67.

[23] Panda, P. and Sahu, G.P. 2012. E-Procurement Implementation: Critical Analysis of the Impact of Success Factors on Project. IUP Journal of Supply Chain Management. 9, 2, 37-65.

[24] Panda, P. and Sahu, G.P. 2013. E-Government Procurement Implementation in India: A cross sectional study. Paper accepted by IJBIS, Inderscience. In press.

[25] Pinto J K, Cleland, D. I., and Slevin, D. P. 2002. The Frontiers of Project Management Research. Project Management Institute, ISBN 1-880410-74-5.

[26] Pinto, J. K., and Slevin, D. 1987. Critical Factors in Successful Project Implementation. IEEE Transactions on Engineering Management, 34, 1, 22-27.

[27] Prabhakar, G.P. 2008. What is Project Success: A Literature Review. International Journal of Business and Management. 3, 9.

[28] Reddick, C.G. 2004. A two stage model of e-government growth: Theories and empirical evidence for US cities. Government Information Quarterly. 21, 1, 51-65.

[29] Rockart, J. F. 1982. The Changing Role of Information System Executive: a Critical Success Factors Perspective. Sloan Management Review. 24, 1, 3-13.

[30] Schoenherr, T., and Tummala, V.M.R. 2007. Electronic procurement: a structured literature review and directions for future research. International Journal of Procurement Management. 1, 1/2, 8-37.

[31] Singh, A. K. 2009. E-Procurement. In R.K. Bagga, and P. Gupta (Eds), Transforming Government, eGovernance Initiatives in India, pp 517-525. Hyderabad, India: ICFAI University Press.

[32] Sundarraj, R.P. and Kumari, K. 2013. Electronic Procurement Systems in India: Importance and Impact on Supply Chain Operations. In J. H. Bookbinder (Ed.), Handbook of Global Logistics (International Series in Operations Research and Management Science: 181).

[33] Tas, B.K.O. and Genis-Gruber, A. 2008. E-Procurement Savings and the Competition Effect: Analysis of Cultural Differences through a Unified Model. Department of Economics Working Paper No: 08-15. Ankara, Turkey: TOBB University of Economics and Technology.

[34] Thai, K.V. 2001. Public Procurement Re-examined. Journal of Public Procurement. 1, 1, 9-50.

[35] Vaidya, K., Sajeev A.S.M., and Callender, G. 2004. E-Procurement Initiatives in the Public Sector: An Investigation into the Critical Success. 13th IPSERA Conference, Catania, Italy. 4-7 April 2004.

[36] Waring, T.S. and Wainwright, D.W. 2008. Issues and Challenges in the Use of Template Analysis: Two Comparative Case Studies from the Field. Electronic Journal of Business Research Methods. 6, 1.

[37] Williams, J.J. and Ramaprasad, A. 1996. A Taxonomy of Critical Success Factors. European Journal of Information Systems. 5, 250-260.

[38] Wisker, G. 2008. The Postgraduate Research Handbook. Basingstoke: Palgrave Mcmillan.